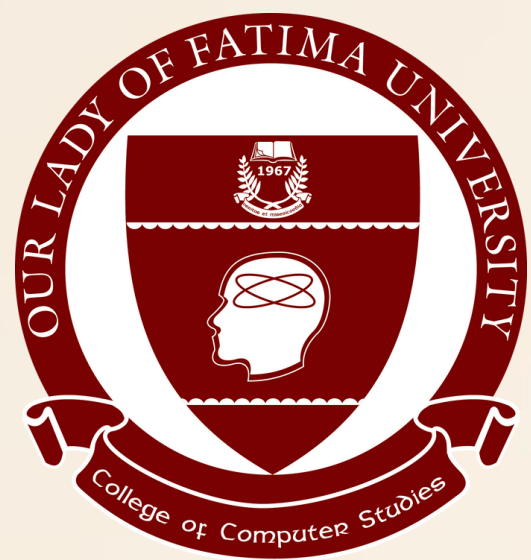




Module 4

Types of Software

ITCL112 - INTRODUCTION TO COMPUTING W/ LAB

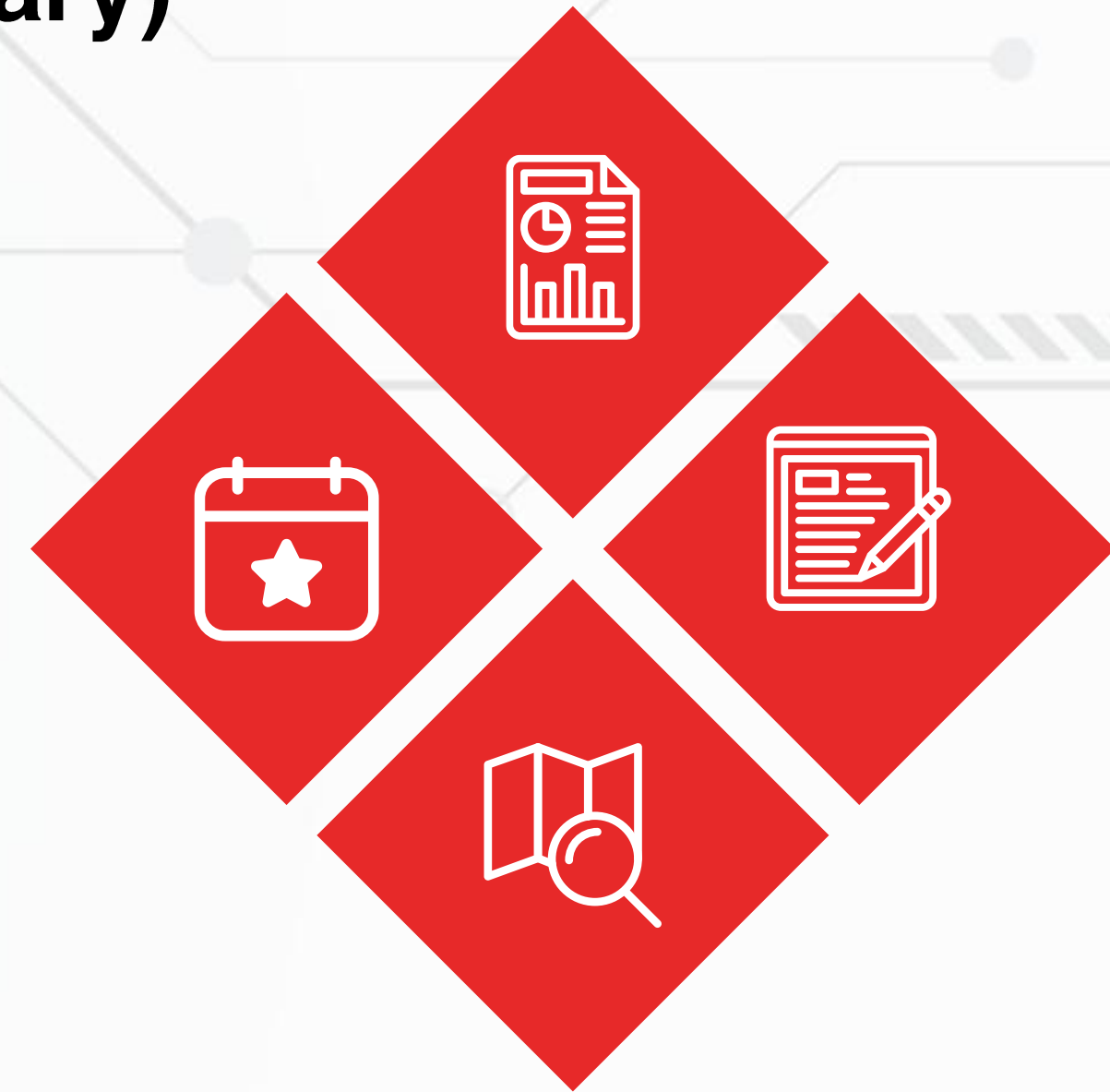
INTRODUCTION



Software is a set of instructions that tells a computer how to perform specific tasks. It is generally divided into two main categories: system software, which manages computer hardware and provides a platform for applications, and application software, which helps users accomplish specific tasks like word processing, browsing, or gaming. Understanding the types of software is essential to know how computers function and how we use them in everyday life.

MODULE 4

- 1. Types of Software**
- 2. Office Productivity Tools (Word, Excel, PowerPoint)**
- 3. Software Licensing (Open Source, Proprietary)**



WHAT IS SOFTWARE?

Software is a set of instructions, programs, or data that tells a computer or electronic device how to work and perform specific tasks. Unlike hardware (the physical parts of a computer), software is intangible it exists as code and instructions stored in memory or storage devices.



SYSTEM SOFTWARE VS APPLICATION SOFTWARE

System Software

Software that manages and controls the computer hardware so other software and users can work with it.

Purpose: Acts as a bridge between the hardware and the user.

Examples:

- Operating Systems (Windows, macOS, Linux, Android, iOS)
- Utility Programs (antivirus, disk management tools, file compression tools)
- Device Drivers (software that makes hardware work, e.g., printer drivers)

Application Software

Software designed for users to perform specific tasks or solve particular problems.

Purpose: Directly helps the user accomplish work, study, entertainment, or daily tasks.

Examples:

- Microsoft Word, Excel, PowerPoint (productivity apps)
- Google Chrome, Firefox (browsers)
- Photoshop (graphics)
- Games, media players

WHAT IS OPERATING SYSTEM?



An Operating System (OS) is the main software that manages computer hardware, software resources, and provides services for computer programs.

It acts as a bridge between the user and the computer hardware, making it possible to run applications smoothly without needing to know how the hardware works.

1. Operating Environment

- An Operating Environment is the setup or platform in which applications run.
- Refers to programs that sit on top of the OS
- It includes the operating system, hardware, and any supporting software needed for programs to work.

2. Programming Damages

- This is likely referring to Programming Errors (or Bugs) that cause problems in software. They are mistakes in coding or logic that lead to program malfunctions.

Types of Error

1. Syntax Error – Mistakes in grammar of the programming language (e.g., missing ;).
2. Logic Error – Program runs but gives wrong results.
3. Runtime Error – Errors that occur while the program is running (e.g., dividing by zero, memory overflow).

WHAT IS OPERATING SYSTEM?



3. Utilities

- Utility programs are special system software designed to help maintain, manage, and optimize the computer.
- They perform tasks that support the operating system.

Example:

- Antivirus software – Protects against malware.
- File compression tools (e.g., WinRAR, 7-Zip).
- Disk cleanup and defragmenter.
- Backup software.

4. Language Processors

A Language Processor is a system program that translates human-readable code (high-level or assembly language) into machine code that the computer can understand.

Types of Language Processors:

- Assembler – Converts Assembly Language → Machine Code.
- Compiler – Translates whole program (C, Java, etc.) into machine code before execution.
- Interpreter – Translates and runs program line by line (e.g., Python).

TYPES APPLICATION SOFTWARE



1. Package Software (Off-the-shelf Software)

Pre-made software developed for general use by many users or organizations.

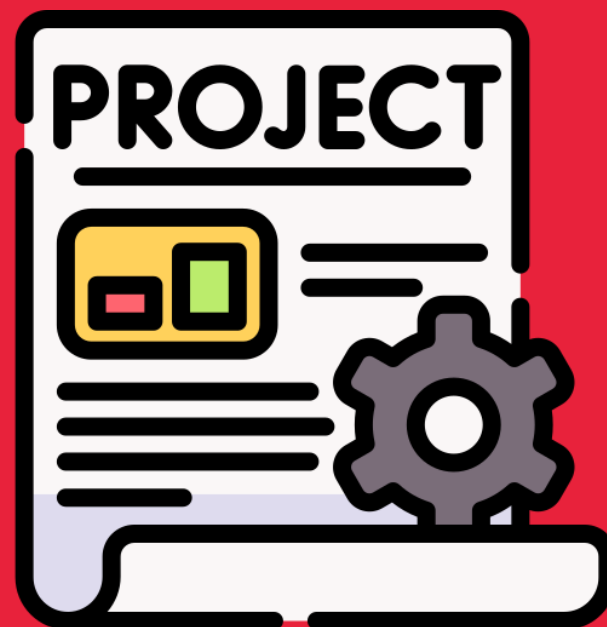
📌 Characteristics:

- Ready-made, can be purchased and installed immediately
- Covers common needs of most users
- Cheaper compared to custom software
- Limited flexibility (cannot be changed much)

📌 Examples:

- Microsoft Office (Word, Excel, PowerPoint)
- Adobe Photoshop
- Google Workspace

TYPES APPLICATION SOFTWARE



2. Custom Program (Tailor-made Software)

Software specially developed for a specific organization or user according to their requirements.



Characteristics:

- Designed to meet unique needs
- More flexible and adaptable
- Expensive and takes time to develop
- Requires maintenance and updates



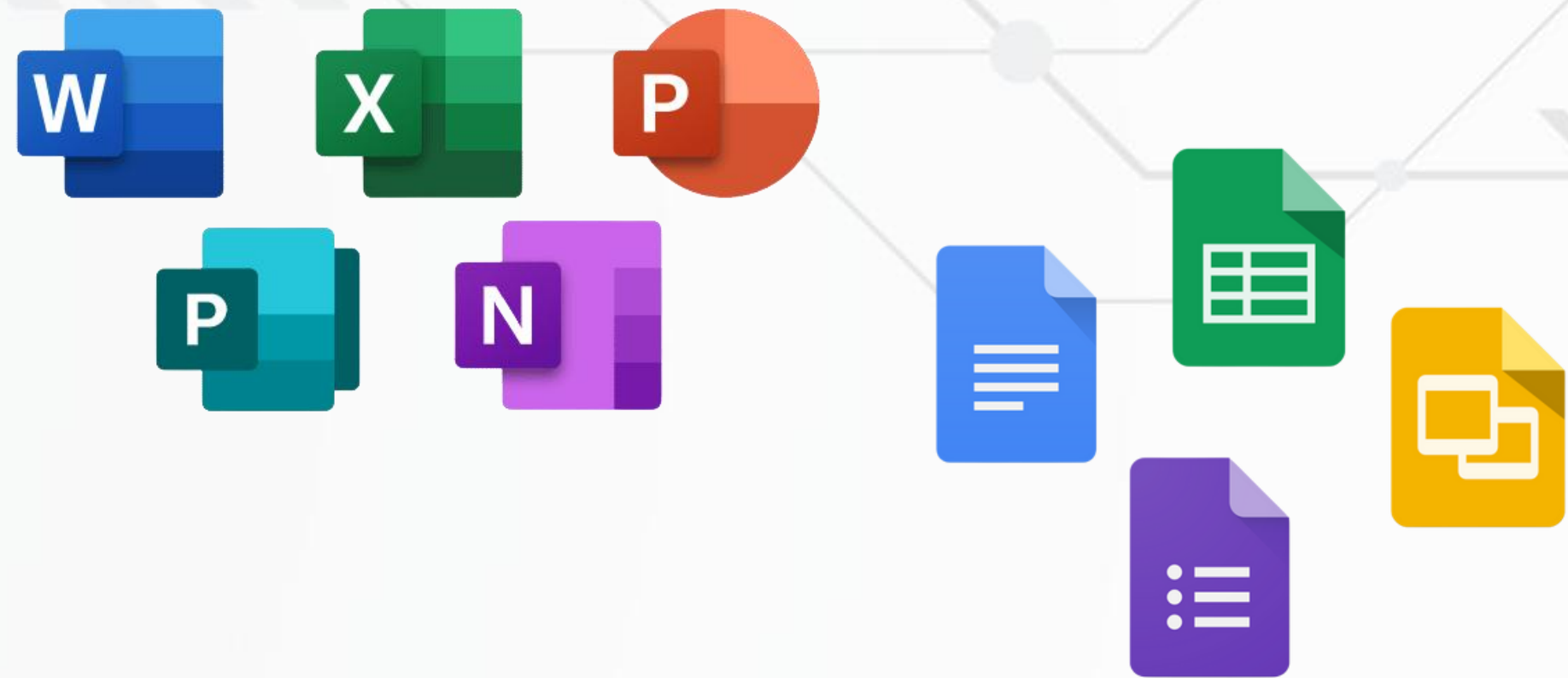
Examples:

- Banking systems made for a specific bank
- Hospital management system for a particular hospital
- Crime Reporting System for Morong Police Station

OFFICE PRODUCTIVITY PACKAGES

Office Productivity Packages are application software programs designed to help users perform everyday tasks efficiently, especially in offices, schools, and businesses. They usually come in a suite (a group of programs) that work together.

- Word Processing Software (e.g., Microsoft Word, Google Docs)
- Spreadsheet Software (e.g., Microsoft Excel, Google Sheets)
- Presentation Software (e.g., Microsoft PowerPoint, Prezi)
- Note-Taking Software (e.g., Evernote, OneNote)



BUSINESS AND ACCOUNTING PACKAGES

Package programs in Business and Accounting are ready-made software designed to handle financial transactions, records, and business operations efficiently. These help organizations in managing accounts, payroll, billing, and decision-making.

- Accounting Software (e.g., QuickBooks, Tally, Sage)
- Payroll Management Software (e.g., ADP, Gusto)
- Enterprise Resource Planning (ERP) Software (e.g., SAP, Oracle ERP)

The logo for Sage, featuring the word "sage" in a green, lowercase, sans-serif font.The logo for Intuit QuickBooks, featuring a green circle with a white "qb" inside, above the word "intuit" in a small, grey, lowercase font, and the word "quickbooks" in a larger, grey, lowercase font.The logo for SAP, featuring the letters "SAP" in a bold, white, sans-serif font, set against a blue background that is a right-angled triangle.The logo for Oracle, featuring the word "ORACLE" in a white, uppercase, sans-serif font, set against a solid red square background.The logo for Gusto, featuring the word "gusto" in a white, lowercase, sans-serif font, set against a solid orange square background.The logo for ADP, featuring the letters "ADP" in a white, stylized, sans-serif font, set against a solid red square background.

WEB BROWSING AND INTERNET UTILITIES

Web browsing is the activity of accessing and navigating information on the World Wide Web (WWW) using a program called a web browser (e.g., Google Chrome, Mozilla Firefox, Microsoft Edge, Safari).

- Web Browsers (e.g., Google Chrome, Mozilla Firefox, Safari)
- Download Managers (e.g., Internet Download Manager - IDM)

Internet utilities are software tools or services that help users manage, maintain, or improve their use of the Internet.

- Email Clients – Gmail, Outlook (for sending/receiving emails).
- File Transfer Tools – FTP clients, Google Drive, Dropbox.
- Security Tools – Firewalls, antivirus, VPNs (for safe browsing).
- Communication Tools – Zoom, Messenger, WhatsApp.
- Search Engines – Google, Bing, Yahoo.



EDUCATIONAL SOFTWARE PACKAGES

Educational Software Packages are computer programs or applications specifically designed to support teaching, learning, and training. They provide interactive and engaging ways for students to learn new concepts, practice skills, and assess their knowledge.

Examples:

- Duolingo (language learning)
- GeoGebra (mathematics and geometry)
- Moodle (e-learning management system)
- Khan Academy (online lessons and exercises)
- Typing Master (keyboard skills training)

coursera



Google Classroom



duolingo

Rosetta Stone 

COMMUNICATION AND COLLABORATION PACKAGES

Communication and Collaboration Packages are types of application software designed to help people exchange information and work together more effectively, especially in organizations, schools, and businesses.

Examples:

- Email software (e.g., Microsoft Outlook, Gmail)
- Messaging apps (e.g., Slack, WhatsApp, Telegram)
- Video conferencing tools (e.g., Zoom, Microsoft Teams, Google Meet)



Hangouts



Google Meet



GAMING AND ENTERTAINMENT PACKAGES

Gaming and Entertainment Packages are bundles of hardware, software, or services designed to provide users with a complete experience for playing games and enjoying multimedia entertainment.

Examples:

- PC & Console Games (e.g., NBA2K23, Fortnite)
- Game Development Software (e.g., Unity, Unreal Engine)
- Media Players (e.g., VLC, Windows Media Player)



WHAT IS SOFTWARE LICENSING

Software Licensing is a legal agreement between the software creator (or vendor) and the user that defines how the software can be used, distributed, or modified.

It acts like a contract that gives the user the right to use the software under specific conditions, while still protecting the developer's intellectual property.

Key Points about Software Licensing:

- 📄 **Defines Usage Rights** – Who can use the software, how many devices, and for what purpose.
- 💻 **Ownership** – The user does not own the software; they only purchase a license to use it.
- 🔒 **Restrictions** – Prevents unauthorized copying, modification, or redistribution.
- 💰 **Payment Terms** – Some licenses require one-time purchases, subscriptions, or royalties.

TYPES OF SOFTWARE LICENSING

Software Licensing is a legal agreement between the software creator (or vendor) and the user that defines how the software can be used, distributed, or modified.

- 1. Proprietary License – Closed-source; users can use but not alter the software (e.g., Microsoft Office).**
- 2. Freeware – Free to use but still copyrighted (e.g., Adobe Acrobat Reader).**
- 3. Shareware – Free trial with limited features, requiring purchase for full use.**
- 4. Opensource License – Source code is available for modification and redistribution (e.g., Linux, Apache).**
- 5. Enterprise/Volume License – Allows organizations to use software on multiple devices under one license.**